

Once a jolly jumbuck...

by Brian Creer
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Being a report of the Type
approval flight tests of
Keith Jarvis' second design
... the J-2

In submitting this report, Brian Creer comments :
I think this is one of the very rare occasions in Australia when a privately built sailplane has been tested under full test conditions and put through a complete test schedule. Perhaps it will encourage others to submit complete test reports instead of *"It-appeared-to-have a good-rate-of-sink-as-I-flew-for-some-time-in-a-20fps-thermal-in-company-with-theBoondywollop-Gliding-Club's-Primary-and-the-difference-in-speeds-and-rate-of-climb-was-not-noticeable.."* type of report.



Three years ago, Adelaide pilot Keith Jarvis first flew his "home-grown" sailplane, Joey – more rightly, the Jarvis J-1. Since then, Joey has flown its way into a place of high respect in the Australian gliding scene.

Its light weight (133 lbs) and small overall size (length 12 ft, span 32 ft) together with its ease of rigging and trailering have brought some envious glances from operators of larger and more unwieldy machines.

Its flying performance, with durations up to seven hours and many worthwhile cross-country flights, and the enthusiastic comments of all who have flown it has encouraged many would-be constructors to seek plans of the little machine.

Because Keith Jarvis is not one to do things by halves or make do when improvements are possible, he decided to get out a really complete set of plans with mods and improvements included.

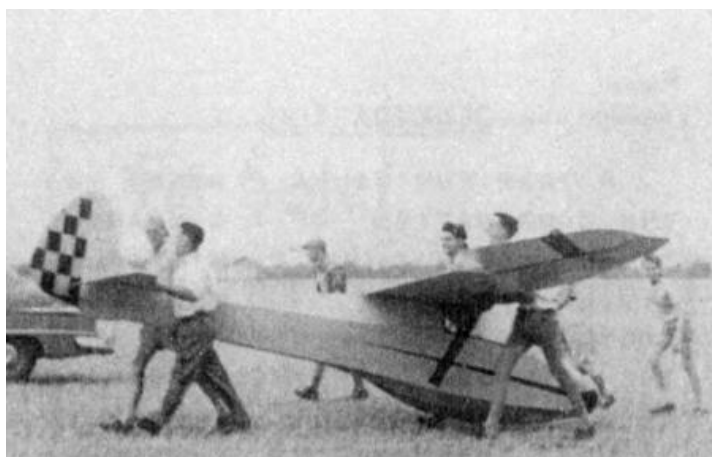
There was another reason too. Keith is not, physically, a large man and Joey's cockpit had been tailored to fit him. He decided to design a larger cockpit and nose for the sailplane while putting it onto paper.

One thing led to another and the old plans were finally thrown away and a new aircraft designed instead. I said Keith doesn't do things by halves! The J-2 was the result. The first of them to be built was completed by Kevin Sedgman of Adelaide and has been named Jumbuck.

Externally, the J1 and J2 are identical but the Jumbuck has a much simplified construction, a new control system, larger cockpit and ... well, look at the comparison table yourself.

I was recently asked by Kevin Sedgman to put Jumbuck through the complete test schedule laid down for the issue of a Certificate of Type Approval.

The flight test was carried out from the Adelaide Soaring Club's base at Gawler and it followed the test schedule used for Type Approval of the ES-52 Kookaburra by DCA pilots. For Jumbuck, however, schedule was amended to the applicable design speeds.



Designer Keith Jarvis (left) and Joey with its original nose (since modified).
Right, the ease of ground handling is apparent as four men lift the Jumbuck and carry it about without effort.

THE TESTS

This test schedule requires the following :

1. A stall from a turn with 30 degrees bank.
 2. A dive to 78 knots (90 mph) during which the main controls shall be exercised gently.
 - 2a. A dive to 68 knots (80 mph) during which ailerons are to be moved to full deflection in both directions. (NOTE: 2 and 2a were repeated with spoilers extended).
 3. Flight with all controls abandoned. Glide trimmed to 48 knots (55 mph), increase and decrease speed by 10 knots and observe time taken to resume level (trimmed) flight.
 4. From a turn with 30 degrees bank, reduce speed until minimum speed for steady turn is reached. Turn through several circles noting (A) minimum turning speed, (B) any control difficulty, (C) any unnatural or reversal effects, (D) the response to small control movements and control position during the turn and (E) time taken to roll out and regain level flight.
 5. The glider shall have safe flying characteristics in the en-route configuration in turns up to 3 ½ G. (Note: these were flown by guesswork as no G-meter was available).
- SPINNING: The machine was also spun from (A) level flight, (B) Medium (slow) turns and steep turns and (C) sideslips.

IN THE AIR

The first few puffs of awakening cumulus were beginning to form to the east of Gawler as I eased myself into Jumbuck's cockpit.

Compared to Joey's cockpit, Jumbuck's is one inch wider, one inch deeper (due to the new control layout) and two inches longer.

From habits acquired from flying Joey, I removed my shoes to ensure full rudder movement. This was forced on me because of the type of shoe I was wearing and because I have such darned large feet.

Anyway, it's nice to fly as per Kronfeld with stockinged feet curled over the rudder pedals.

The first point I noticed when easing my bottom onto its resting place was Kebin Sedgeman's "hammock-type" seat. This is an extremely comfortable thing and I don't doubt more will be found in sailplanes in the future.

Resting comfortably in the "hammock", I studied the panel of the Jumbuck. From the left, it runs : ASI (in knots), Cook compass, variometer, a pressure operated turn and slip indicator, altimeter and outside air temperature gauge.

Below the air temp. gauge on the lower right of the panel are three knobs – one green, one red and below them a black one.

The green knob releases a 4-ft diameter braking-cum-anti-spin parachute hosed below the rudder. Pull the red knob to jettison the 'chute.

The black knob is pumped to build up pressure to run the turn and slip indicator for a few minutes after which it is pumped again.

The towline release is on the floor just left of the control column and the spoiler knob is below the right elbow in a position guaranteed to cure the most advanced case of rheumatism after three tugs.

The first flight, lasting 40 minutes, began from an auto tow with Jumbuck unsticking after a long run. The length of run is characteristic of the Joey too, and is due in part to the low ground angle of both machines. In this case lack of wind also contributed.

At 500 feet altitude I ran through a patch of well-defined lift (the only sort) so I released and turned back into it.

Circling in the lift and travelling smoothly in the direction of UP it was soon apparent that, although the elevator and ailerons were well harmonised, the rudder control was a bit fierce by comparison with the average sailplane. Keith Jarvis intends to cure the sensitivity of the rudder by gearing it down slightly.

The thermal took us to 2900 feet and I then began the test schedule. In a medium turn the speed can be pulled back to 30 knots before a reverse stick pressure warns of the approaching stall.

If speed is further reduced, this reversal is immediately followed by the dropping of the noses and wing and a spin begins. Recovery may be effected immediately the nose drops.

Ailerons are particularly good in thermals and the whole aircraft is light enough to "feel" lift. Rate of roll is exceptional, as can be imagined with such a small span. However, some degree of adverse yaw is apparent.

PENETRATION

A trap for young players is the combination of a comparatively low permissible diving speed of 100 mph (78 knots for test purposes) and the general airframe cleanliness.

Once the nose is lowered, even slightly, the ASI winds up very quickly and to exceed the design diving speed would not be difficult.

This is encouraged by the similarity of sound at all speeds. The Jumbuck sounds basically the same at 40 knots as it does at 70 knots, but this might be merely a matter of getting used to the craft.

Vision over the low nose is good in all directions even for a pilot slightly shorter than my 5' 10 1/2".

On the second launch (downwind take-off this time) and after a climb from 400 feet to 2700 feet, I continued to explore Jumbuck.

I found its spinning characteristics interesting but not dangerous. It entered a spin cleanly from level flight and I let it spin through three complete turns in a fairly nose-down attitude.

The airspeed did not fluctuate from an indicated 40 knots and recovery was immediate when opposite rudder was applied and the stick centralised.

The second spin from level position was begun, but this time the controls were fully crossed. This move revealed that after one turn, the spin became flat and skidding and the airspeed fluctuated wildly between 40 and 60 knots.

Several spins were then made with and without crossed controls and in each case when the controls were crossed the spin became flat and the air-speed went wild.

Similar results came from spins from turns and sideslips. In the latter case, control reversal was evident early during the slip (at about 40 knots indicated) and increased until the machine flick-rolled out and began spinning.

There is such a pronounced reversal, and it is apparent so long before anything violent happens, that this could be regarded as a really goofy stall-warning indicator.

NO JOY FOR FAT BOYS

The hands-off test had to be abandoned when I found that with my 12 stone (168 lbs) on board, Jumbuck would not fly hands-off at less than 60 knots.

I would suggest here that Kevin fit a trimming device – perhaps a spring-loaded one such as now fitted to the Joey.

Kevin had placed the spoiler control on the right side of the cockpit to suit his own taste. Personally, I found it impossible to reach with the right hand and so awkward to use, cross-over fashion, with my left that I gave up and made spoiler-less approaches.

On my last approach, I popped the tail-chute and its braking effect was immediate and considerable. At present. The 4-ft diameter 'chute has no vent hole in the centre and it tends to oscillate rather badly,

This oscillation was noticed on the final approach by small sideways flicks of the nose. However, the cure is simple and Kevin is already sharpening his scissors.

The value of the tail 'chute for braking was most apparent when, at a height of about 4 feet, I jettisoned it. Jumbuck shot ahead like a greyhound to touch down a good hundred yards further on.

Normally the 'chute is not popped for landing unless it is necessary to halt in a very small space. Also, it is not normally jettisoned but I did so because I wanted a comparison of braked and unbraked performance and also because I did not want the 'chute to get damaged during the landing.

SUMMING UP

Design: Sensible, thorough and conventional.

Construction: Workmanlike, with attention to detail.

Appearance: (Male) rugged, clean and light, yet strong. (Female) cute and cuddly.

Comfort: Average to good for average-sized people – ideal for those under 150lbs and 5'8".

Performance: Min sink 2.5 fps at 35 knots, best glide ratio 23 to 1 at 37 knots. Quite fair enough.

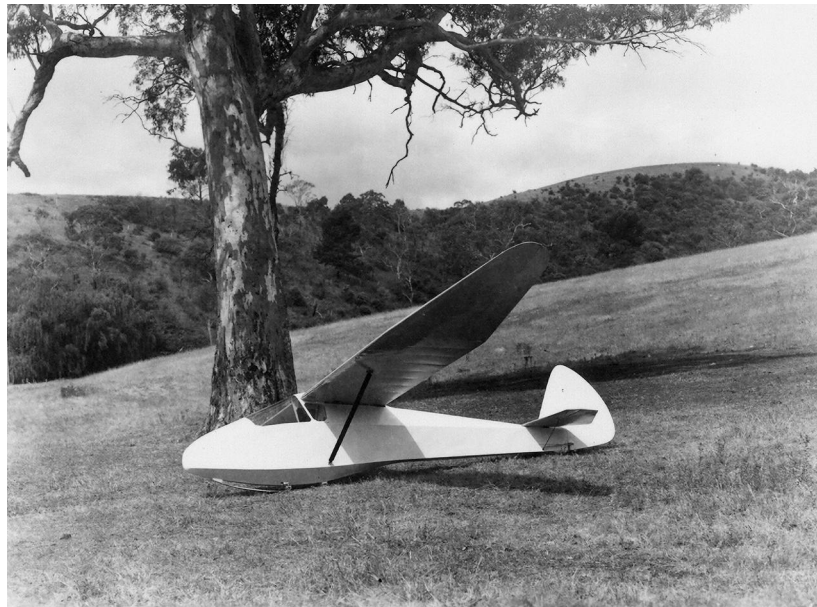
Controls: Well harmonised except for the rudder. Statically, the ailerons are a bit stiff. Spoiler controls are great for Bug-Eyed Monsters and Octopuses. Mild torture for Homo Sapiens.

Flight characteristics: In many ways they are like the characteristics of a good sports car – mannerly at all times but making it easy to exceed the speed limit. Aerobatically fine doing loops, stall-turns, roll off the top, barrel-rolls etc.

Personal desirability: Extremely high.

COMPARISON TABLE

	J-1 Joey	J-2 Jumbuck
Empty weight	133 lbs	140 lbs
Design load	150 lbs	170 lbs
Wing span	32 feet	32 feet
Wing area	80 sq. ft.	80 sq. ft.
Wing loading	3.54 ppsf	3.87 ppsf
Stalling speed	30 mph	31 mph
Category	semi-aerobatic	semi-aerobatic



The Joey and Jumbuck are externally similar but have a few structural differences.